

BuildLink

AI-Powered Construction Ecosystem for Pakistan

Voice AI | Floor Plan Generation | Smart Matching | Escrow Payments | 7 AI Features

Team BuildLink

AGENDA

What we'll cover

01	The Problem	Why building a home in Pakistan is broken
02	Our Solution	BuildLink's end-to-end AI-powered platform
03	AI Features	7 Gemini-powered intelligence systems
04	Voice AI	Urdu/English voice input for accessibility
05	Platform Features	Escrow, milestones, deliverables, ratings
06	Design Timeline	8 sequential design milestones
07	User Roles	Customer, Engineer, Admin workflows
08	Tech Stack	React, Node, MongoDB, Gemini
09	Impact & Vision	Why this matters for Pakistan

THE PROBLEM

Building a home in Pakistan is broken

Construction is the 2nd largest unorganized sector. Homeowners face a fragmented, opaque, and intimidating process.



No Cost Transparency

Homeowners have zero idea what construction costs until committed. Budgets overrun 30-50%.



Finding Engineers is Hard

No verified marketplace. Word-of-mouth only.
No ratings, no credentials, no accountability.



No Trust or Process

No escrow, no milestones, no revision tracking.
Disputes common, no recourse for either party.

300,000+ houses/year | \$15B+ industry | ZERO dominant digital platform

OUR SOLUTION

One platform, every step -- powered by AI

From describing a project in plain Urdu to getting an AI floor plan, matched engineer, escrow, and milestone delivery.



Key Differentiator: First construction platform where you can create a project entirely by voice in Urdu.
Designed for Pakistan's 42% illiterate population -- no typing required at any step.

AI FEATURES | GOOGLE GEMINI

7 AI systems, one platform

V

Voice-to-Project

Speak in Urdu/English -- AI parses into full project brief

\$

NLP Cost Estimator

Plain text to localized PKR cost breakdown with confidence

R

Risk Analysis

Structural, regulatory, budget risks with severity ratings

C

AI Construction Chatbot

Ask any question -- AI answers with project-aware context

F

AI Floor Plans

Room layouts with doors, windows. Amend via natural language

M

Smart Engineer Matching

ML scoring by specialization, location, experience, fit

B

Material Breakdown

AI-generated bill of quantities -- materials, costs, amounts

ACCESSIBILITY-FIRST DESIGN

Voice AI -- built for Pakistan

42% of Pakistan's population is illiterate. BuildLink lets users create projects, chat, and negotiate entirely by voice.

U Urdu Voice Input

Full Urdu speech recognition (ur-PK locale).
Say your requirements in Urdu and
AI creates the entire project brief.

V Voice-to-Project

Describe your dream home by speaking.
AI extracts title, type, rooms, floors,
plot size, budget, location, and style.

C Voice Messaging

Mic button on every chat input.
Speak messages in Urdu -- transcribed
and sent. No keyboard needed.

R Voice Rate Negotiation

Say 'paanch lakh' and the field fills
with 500,000. Negotiate construction
rates entirely by voice.

PLATFORM FEATURES

End-to-end construction management

01 Escrow Payments

Funds held in escrow, auto-released per milestone approval. Both parties protected.

02 Sequential Milestones

8 design-phase milestones. Each must be approved before next starts. Full revision history.

03 Deliverables System

Engineers upload files per milestone. Submit for customer review. Approve / revise / reject

04 Verified Engineers

License verification, specialization checks, admin approval. Ratings and project history.

05 Chat & Negotiation

Built-in messaging with voice input. Rate proposals, counter-offers, acceptance in one thre

06 Ratings & Reviews

Both parties rate after completion. Ratings update engineer profiles, building marketplace trust.

07 Admin Dashboard

User management, engineer verification, project oversight. Full control over platform ecosy

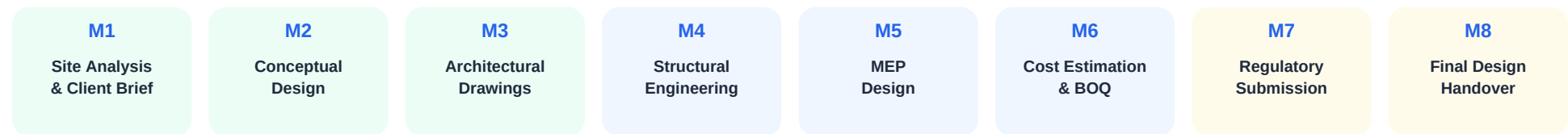
08 Floor Plan Export

Download AI floor plans as high-res PNG (3x) or SVG vector files. Doors, windows, labels.

MILESTONE SYSTEM

8 design milestones, sequential delivery

Each milestone: completed > submitted > approved before the next can start. Customers approve, revise, or reject.



● Approve (releases escrow) ● Request Revision (with comment) ● Reject (engineer notified) ● In Progress

Engineer: Start Milestone > Upload Deliverables > Submit for Review > Customer Decides > Next

Customer: View Deliverables > Approve / Revise (with comment) / Reject > Escrow auto-releases

USER ROLES & FLOWS

Three user types, one ecosystem

Customer

- > Create project (voice or text)
- > Get AI cost estimate
- > Browse / get matched engineers
- > Select engineer, fund escrow
- > Track milestones & deliverables
- > Approve / revise / reject work
- > Rate engineer after completion

Engineer

- > Browse available projects
- > Apply with proposal & budget
- > Accept/decline when selected
- > Negotiate rate via chat (voice)
- > Start milestones sequentially
- > Upload deliverables, submit
- > Get paid via escrow release

Admin

- > Verify engineer licenses
- > Manage all users
- > Oversee all projects
- > Toggle user access
- > Delete problematic content
- > Platform analytics dashboard

TECHNOLOGY

Tech stack

React

Frontend SPA

Component-based UI,
React Router, Context API**Node + Express**

REST API

JWT auth, Multer uploads,
Morgan logging, CORS**MongoDB**

Database

Mongoose ODM,
complex schemas, indexing**Google Gemini**

AI Engine

7 AI features via
@google/genai SDK**Web Speech API**

Voice Recognition

ur-PK + en-US locales,
real-time transcription**JWT + bcrypt**

Authentication

Token-based auth,
password hashing, RBAC**SVG Renderer**

Floor Plans

Dynamic SVG generation,
PNG/SVG export (3x res)**Multer**

File Uploads

Deliverable management
per milestone tracking

THANK YOU

BuildLink

AI-Powered Construction Ecosystem for Pakistan

Voice AI (Urdu) | AI Floor Plans | Escrow Payments | 8 Milestones | 7 AI Features

Team BuildLink | Al-Khidmat Foundation AI Hackathon 2026

React | Node.js | MongoDB | Google Gemini | Web Speech API